Comparison of Episcleral Plaque and Proton Beam Radiation Therapy for the Treatment of Choroidal Melanoma

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Purpose: To compare the efficacy of iodine-125 (^{125}I) and ruthenium-106 (^{106}Ru) episcleral plaque radiation therapy and proton beam radiation therapy (PBRT) in the treatment of choroidal melanoma.

Design: A retrospective, nonrandomized comparative study.

Methods: Charts of patients treated with ^{125}I, ^{106}Ru, and PBRT for choroidal melanoma between January 1988 and June 1996 at St. Bartholomew's Hospital and Moorfields Eye Hospitals were reviewed.

Main Outcome Measure: Local control of choroidal melanomas after ^{125}I, ^{106}Ru, or PBRT.

Results: A total of 597 patients were identified (^{125}I = 190, ^{106}Ru = 140, PBRT = 267). Patients treated with ^{106}Ru had a significantly greater risk of local tumor recurrence than did patients treated with either ^{125}I (P = 0.0133; confidence interval [CI], 1.26–7.02; risk ratio, 2.97) or PBRT (P = 0.0097; CI, 1.30–6.66; risk ratio, 2.94). A stepwise Cox proportional hazard model found maximal basal diameter to be a significant covariate (P = 0.0033).

Conclusion: Patients treated with ^{106}Ru had a significantly greater risk of local tumor recurrence than did those patients treated with either ^{125}I or PBRT. *Ophthalmology 1999;106:1579–1587*

Episcleral plaque and charged particle radiation therapy are accepted treatments for the management of choroidal melanomas. Reports of radiation for the treatment of choroidal melanoma date to 1930. Moore[1] implanted radon seeds in what was then termed *choroidal sarcoma*. Stallard[2] later pioneered the use of episcleral plaque radiation therapy in the treatment of choroidal melanoma using cobalt-60. Tumors received 100 Gy of irradiation to their apex. Radiation was hypothesized to destroy the reproductive integrity of choroidal melanomas, effectively sterilizing them.[3,4] This conservative therapy provided effective local tumor control and allowed retention of the eye.

Since Moore and Stallard's pioneering efforts with radon and cobalt, other forms of radiation therapy have been used successfully. Episcleral plaque radiation therapy has used several different isotopes: iodine-125 (^{125}I), ruthenium-106 (^{106}Ru), iridium-192, and palladium-103.[5–15] All have been reported to provide effective local tumor control. Charged particles in the form of protons and helium ions also have been reported to provide effective local tumor control.[16–20] Microwaves and gamma knife radiation surgery have been used to treat choroidal melanomas.[21–24] The purpose of this study is to compare the efficacy of ^{125}I and ^{106}Ru episcleral plaque radiation therapy and proton beam radiation therapy (PBRT) in the local treatment of choroidal melanoma.

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Patients and Methods

A retrospective chart review of patients treated with ^{125}I, ^{106}Ru, and PBRT for choroidal melanoma from January 1988 through June 1996 at St. Bartholomew's Hospital and Moorfields Eye Hospital, London, England, was performed. The following clinical data were extracted from the patients' medical records: treatment type, age, gender, affected eye, tumor location, tumor height and maximal basal diameter, presence of subretinal fluid, pretreatment visual acuity, visual acuity at last examination, and months of follow-up. The development of metastatic disease and patient mortality were also recorded.

All patients were examined before and after radiation by one investigator (JLH). Fundus photography and diagnostic ultrasound were performed during each examination. Outcomes were measured in terms of local tumor control. Tumor growth after radiation therapy was considered a treatment failure. Tumor growth was documented by clinical examination, fundus photography, and serial ultrasounds. Treatment endpoints were tumor growth, enucleation, and patient death. Follow-up was calculated in months from the time of treatment to tumor growth, enucleation, or last visit. Patients receiving more than one treatment were included only once within this study with their initial cohort and deemed treatment failures.

Patients treated with episcleral plaque radiation therapy received 100 cobalt Gray Equivalents to the tumor apex. Dosimetry was maintained above 45 cGy/hr during treatment. Only patients who had undergone plaque therapy treated with either ^{125}I or ^{106}Ru were included in this study. Both indirect ophthalmoscopy and transillumination were used to demarcate tumor margins intraoperatively. Dummy plaques were placed initially using 5–0